

Lecture 10: Stochastic Calculus

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1 Stochastic Integral vs. Payoff of a Trading Strategy

We consider the Bachelier model for stock price dynamics. Let us fix a time-horizon $T > 0$, consider the following trading strategy: at any time $t \in [0, T]$, we hold $\Delta(t)$ shares of stocks. The process $\{\Delta(t)\}_{t \in [0, T]}$ is adapted to the filtration $\{\mathcal{F}_t\}_{t \in [0, T]}$, since we can adjust our holdings based on what we have seen so far. In an infinitesimal time interval $[t, t + dt]$, the value change of this portfolio is given by

$$\Delta(t)(W(t + dt) - W(t)).$$

Overall, the value change of this portfolio from time 0 to time T , without considering interest rate, is given by

$$I(T) = \int_0^T \Delta(t) dW(t). \quad (1)$$

Expressions like the one in (1) is called stochastic integrals. It is different from classical Lebesgue/Riemann integrals, for which the differential $dW(t)$ can be expressed with dt . We will formally give an interpretation of (1) for simple integrand $\{\Delta(t)\}_{t \in [0, T]}$. With this definition we study two important properties of stochastic integrals. Finally, we allow the integrand to be an adapted stochastic process to we give a non-trivial example which distinguish stochastic calculus with classical calculus.

2 Construction of the Integral

Let $\Pi = \{t_0, t_1, \dots, t_N\}$ be a partition of the interval $[0, T]$:

$$0 = t_0 < t_1 < \dots < t_N = T.$$

We consider trading strategies that are constants on interval $[t_{i-1}, t_i)$, for $1 \leq i \leq N$. These strategies correspond to discrete re-balance. The position of stocks only changes at time t_i . Then for any given $t \in [0, T]$, there exists a unique k such that $t \in [t_k, t_{k+1})$. The value change of the portfolio from time 0 to time t is thus given by

$$I(t) = \sum_{i=1}^k \Delta(t_{i-1})(W(t_i) - W(t_{i-1})) + \Delta(t_k)(W(t) - W(t_k)). \quad (2)$$

The above expression is called the Itô integral of the simple process $\Delta(t)$. We can write it as

$$I(t) = \int_0^t \Delta(u) dW(u). \quad (3)$$

3 Properties of Itô Integral

Since Brownian motion $\{W(t)\}_{t \geq 0}$ is a martingale, it can be shown that the process given by Itô integral (2), $\{I(t)\}_{t \in [0, T]}$, is also a martingale.

Theorem 1. *The Itô integral defined by (2) is a martingale.*

We can prove this property directly by showing that $E(I(t) - I(s)|\mathcal{F}_s) = 0$ for any $0 \leq s < t \leq T$. If s, t are in the same interval $[t_k, t_{k+1})$, then from (2) we know that

$$I(t) - I(s) = \Delta(t_k)(W(t) - W(s)). \quad (4)$$

Since $\Delta(t_k)$ is $\mathcal{F}_s$ -measurable¹, using martingale property of Brownian motion we have

$$E(I(t) - I(s)|\mathcal{F}_s) = \Delta(t_k)E(W(t) - W(s)|\mathcal{F}_s) = 0. \quad (5)$$

If s, t are in different interval, say, $s \in [t_l, t_{l+1})$ and $t \in [t_k, t_{k+1})$, for some $l < k$. Then using (2) we know that

$$I(t) - I(s) = \Delta(t_k)(W(t) - W(t_k)) + \sum_{l+2 \leq i \leq k} \Delta(t_{i-1})(W(t_i) - W(t_{i-1})) + \Delta(t_l)(W(t_{l+1}) - W(s)). \quad (6)$$

We will use the tower property of conditional expectation:

$$E(I(t) - I(s)|\mathcal{F}_s) = E(E(I(t) - I(s)|\mathcal{F}_{t_i})|\mathcal{F}_s), \quad i \geq l+1. \quad (7)$$

From (6) we know that

$$E(I(t) - I(s)|\mathcal{F}_{t_k}) = \sum_{l+2 \leq i \leq k} \Delta(t_{i-1})(W(t_i) - W(t_{i-1})) + \Delta(t_l)(W(t_{l+1}) - W(s)). \quad (8)$$

By induction, every time we take the conditional expectation given $\mathcal{F}_{t_i}$, we can get rid of the term involving $\Delta(t_i)$, for any $l+1 \leq i \leq k$. Using tower property (7) we have

$$E(I(t) - I(s)|\mathcal{F}_s) = E(\Delta(t_l)(W(t_{l+1}) - W(s))|\mathcal{F}_s) = \Delta(t_l)E(W(t_{l+1}) - W(s)|\mathcal{F}_s) = 0. \quad (9)$$

Because $I(0) = 0$, it follows that $E(I(t)) = 0$ for any $t \in [0, T]$. As a result, $\text{Var} I(t) = EI^2(t)$, a quantity that can be evaluated by the formula in the next result.

Theorem 2 (Itô isometry). *The Itô integral defined in (2) satisfies*

$$EI^2(t) = E \int_0^t \Delta^2(u) du. \quad (10)$$

We simplify notation by setting $D_i = W(t_i) - W(t_{i-1})$ for $1 \leq i \leq k$ and $D_{k+1} = W(t) - W(t_k)$. Then $I(t) = \sum_{i=1}^{k+1} \Delta(t_{i-1})D_i$, and

$$I^2(t) = \sum_{i=1}^{k+1} \Delta^2(t_{i-1})D_i^2 + 2 \sum_{1 \leq i < j \leq k+1} \Delta(t_{i-1})\Delta(t_{j-1})D_i D_j. \quad (11)$$

Note that for $i < j$, $\Delta(t_{i-1})\Delta(t_{j-1})D_i$ is $\mathcal{F}_{t_{j-1}}$ -measurable, and D_j is independent of $\mathcal{F}_{t_{j-1}}$ and has zero mean, so

$$E(\Delta(t_{i-1})\Delta(t_{j-1})D_i D_j) = E(\Delta(t_{i-1})\Delta(t_{j-1})D_i E(D_j|\mathcal{F}_{t_{j-1}})) = 0. \quad (12)$$

Hence,

$$\begin{aligned} EI^2(t) &= \sum_{i=1}^{k+1} E(\Delta^2(t_{i-1})D_i^2) = \sum_{i=1}^{k+1} E(\Delta^2(t_{i-1})E(D_i^2|\mathcal{F}_{t_{i-1}})) \\ &= \sum_{i=1}^k E\Delta^2(t_{i-1})(t_i - t_{i-1}) + E\Delta^2(t_k)(t - t_k) = E \int_0^t \Delta^2(u) du. \end{aligned} \quad (13)$$

From Theorem 2 we see that the variance of the random variable $I(t)$ is the expectation of $\int_0^t \Delta^2(u) du$, so what is the integral itself? It turns out to be the quadratic variation. We give an informal proof of this fact. Let $\langle I \rangle(t)$ be the quadratic variation at time 0, then

$$\langle I \rangle(t) = \int_0^t (dI(t))^2 = \int_0^t (\Delta(u)dW(u))^2 = \int_0^t \Delta^2(u)(dW(u))^2 = \int_0^t \Delta^2(u) du. \quad (14)$$

¹In other words, its value is already known at time s .

4 Itô integrals for General Integrands

We can extend Itô integrals to any $\mathcal{F}_t$ adapted stochastic process $\Delta(t)$ which is right continuous and has left limit. Such integrand corresponds to a trading strategy that involves continuous subtle re-balance from time to time, and sometimes abrupt big changes in holdings. If we decide to abruptly change our position at time t , then our holdings changes suddenly at time t , but the holdings does not change before t .

The nature way to define Itô integrals for this kind of integrands is to use simple integrand to approximate. If $\{\Delta(u)\}_{u \in [0, T]}$ is square-integrable:

$$E \int_0^T \Delta^2(u) du < \infty, \quad (15)$$

then we can always use this approximation to define the integral of $\{\Delta(u)\}_{u \in [0, T]}$ with respect to Brownian motion:

$$I(t) = \int_0^t \Delta(u) dW(u) := \lim_{n \rightarrow \infty} \int_0^t \Delta_n(u) dW(u), \quad (16)$$

where $\Delta_n(t)$ is a piecewise constant function of t that approximates $\Delta(t)$.

Since Itô integrals for general integrands are limits of Itô integrals for simple integrands, general Itô integrals also have the following property:

- As a function of t , the integral $I(t)$ in (16) is continuous.
- For each $t \in [0, T]$, $I(t)$ is $\mathcal{F}_t$ -measurable.
- Linearity: sum of integrals is equal to integrals of sum.
- $\{I(t)\}_{t \in [0, T]}$ is a martingale.
- Itô isometry: $E I^2(t) = E \int_0^t \Delta^2(u) du$.
- Quadratic variation $\langle I \rangle(t) = \int_0^t \Delta^2(u) du$.

Example 1. We compute $\int_0^t W(u) dW(u)$ using definition (16). Let us fix $t > 0$, choose a large integer N and consider the partition $\{0, \frac{t}{N}, \frac{2t}{N}, \dots, t\}$. We define a simple process

$$\Delta_n(t) = \begin{cases} W(0) = 0 & \text{if } t \in [0, \frac{t}{N}), \\ W(\frac{t}{N}) & \text{if } t \in [\frac{t}{N}, \frac{2t}{N}), \\ \vdots & \\ W(\frac{(N-1)t}{N}) & \text{if } t \in [\frac{(N-1)t}{N}, t) \end{cases}. \quad (17)$$

To simplify notations we denote $W_i = W(\frac{it}{N})$, for $0 \leq i \leq N$. Then

$$\begin{aligned} \int_0^t \Delta_n(u) dW(u) &= \sum_{i=0}^{N-1} W_i (W_{i+1} - W_i) = \sum_{i=0}^{N-1} \frac{W_{i+1} + W_i}{2} (W_{i+1} - W_i) - \sum_{i=0}^{N-1} \frac{W_{i+1} - W_i}{2} (W_{i+1} - W_i) \\ &= \frac{1}{2} \sum_{i=0}^{N-1} \{W_{i+1}^2 - W_i^2\} - \frac{1}{2} \sum_{i=0}^{N-1} \{W_{i+1} - W_i\}^2 = \frac{1}{2} \{W_N^2 - W_0^2\} - \frac{1}{2} \sum_{i=0}^{N-1} \{W_{i+1} - W_i\}^2 \\ &\rightarrow \frac{1}{2} W^2(t) - \frac{1}{2} t. \end{aligned} \quad (18)$$

Thus, we have

$$\int_0^t W(u) dW(u) = \frac{1}{2} W(t)^2 - \frac{1}{2} t \neq \frac{1}{2} W^2(t)!!! \quad (19)$$